TELESCOPE

Project Planning & Management Blueprint — Proposal, Architecture & Roadmap

DOMAIN	CAPABILITIES	PROVIDERS	PLANNING WINDOW	VERSION
Egyptian Telecom Market Intelligence	Sentiment Analysis & Hybrid Recommendations	WE, Orange, Vodafone, Etisalat	26 Sep – 4 Dec 2026 (10 Weeks)	v1.0, 27 Oct 2026

1. Project Proposal & Overview

Problem. Egyptian telecom customers compare providers and offers using scattered information: app store reviews, subjective personal experiences, and dynamic promotional descriptions. Without a unified analytical pipeline, comparing these heterogeneous sources consistently remains challenging for market analysts and consumers.

Solution. TeleScope is a robust, data-driven platform that unifies Google Play store reviews, curated survey responses, and structured offer data for the four major Egyptian telecom providers (WE, Orange, Vodafone, and Etisalat). It evaluates customer sentiment, benchmarks provider performance, synthesizes offer characteristics, and delivers personalized recommendations via an intuitive Streamlit interface.

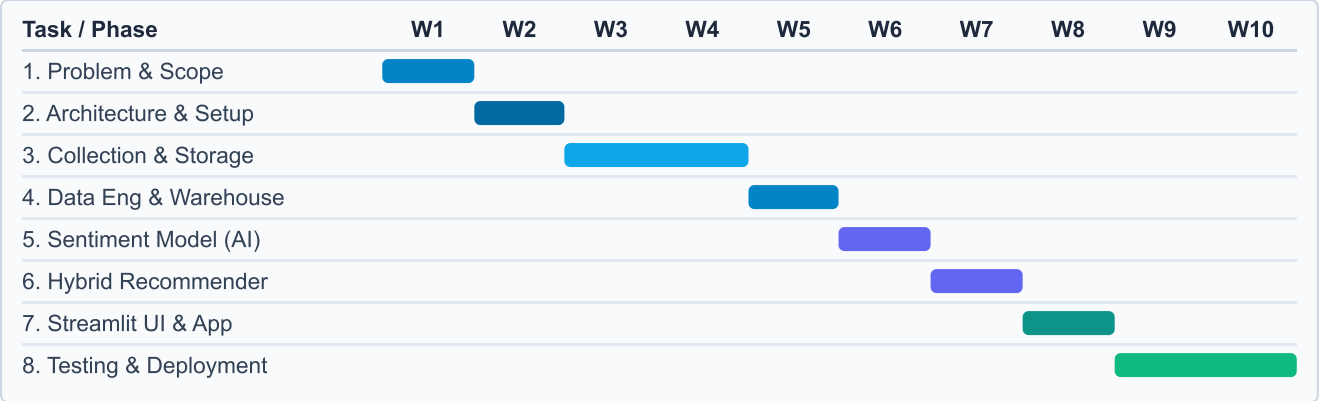
1.2 Objectives & Core Scope

Overall Objective. Design, engineer, and deploy TeleScope to decode Egyptian telecom market sentiment and simplify provider and offer selection using multi-source data processing, sentiment modeling, and hybrid recommendations.

In Scope	Out of Scope
• Google Play reviews for WE, Orange, Vodafone, Etisalat • Primary user survey responses on telecom preferences • Telecom offer records, pricing, and package attributes • End-to-end data pipelines (Airflow, Azure Blob, PostgreSQL) • Sentiment analysis models and hybrid recommender logic • Streamlit UI dashboards and comparison views	• Real-time data streaming as a mandatory core requirement • Native mobile applications (Android / iOS) • Direct integration with providers' internal billing systems • Training large foundational language models from scratch • Live network coverage hardware testing or physical speed probes

2. Project Plan & Gantt Chart Timeline (10-Week Window)

The project follows a structured 10-week schedule from September 26 to December 4, 2026, visualized below as a Gantt chart tracking phases from data engineering setup to final deployment.



3. Task Assignment & Team Roles (6 Members)

Team responsibilities are clearly partitioned to leverage individual expertise while ensuring collaborative mastery of core data engineering principles.

Phase 1: Data Engineering (All 6 Members)

Team: All 6 members (Ahmed Ehab, Mahmoud Eid, Mostafa Abdelkader, Mostafa Abbas, Ramadan Hisham, Nour El-Ein Mohamed).
Responsibility: Collaborative development where every member builds and owns a complete end-to-end data pipeline from scrapers to Azure Blob and PostgreSQL.

Phase 2: Data Analytics

Team: Ahmed Ehab, Mahmoud Eid.
Responsibility: Exploratory data analysis, market trend comparisons, survey insight generation, and visualization support for dashboards.

Phase 2: Artificial Intelligence (AI)

Team: Mostafa Abdelkader, Mostafa Abbas.
Responsibility: Sentiment analysis model development, feature engineering, and the hybrid recommendation ranking algorithms.

Phase 2: Software Development (App / UI)

Team: Mostafa Abdelkader, Ramadan Hisham, Nour El-Ein Mohamed.
Responsibility: Streamlit application architecture, services layer integration, responsive UI design, and user journey testing.

4. Risk Assessment & Mitigation Plan

Proactive identification of technical, operational, and scheduling risks ensures timely project delivery and graceful fallback handling.

Identified Risk	Level	Mitigation Strategy & Solution
Incomplete Data Collection / Limits	Med / High	Monitor ingestion counts per provider and source; implement safe retry logic and document collection gaps transparently.
Data Imbalance Across Providers	High	Track language and sample distributions; explicitly report data coverage limits in analytical summaries to avoid bias.
Azure Blob Storage / Network Outage	Medium	Test storage connectivity early; maintain local fallback caching and document recovery procedures.
Weak Sentiment Model Performance	Medium	Establish clear baseline metrics; inspect language-specific classification errors and refine preprocessing rules.
Schedule Compression & Integration Delays	High	Agree on interface contracts during Week 2; prioritize core end-to-end functionality and protect Week 10 for stabilization.

5. Key Performance Indicators (KPIs)

Project success and operational excellence are measured against quantitative technical and user-centric benchmarks throughout development and deployment.

KPI Category	Target Metric	Measurement & Purpose
System Performance	API / Dashboard Response Time < 1.5 seconds	Ensures fluid navigation and snappy interaction across Streamlit analytical views.
Platform Reliability	System Uptime > 98.5% during evaluation window	Validates pipeline orchestration robustness and stable application deployment.
User Adoption & Utility	User Satisfaction & Task Completion Rate > 85%	Measured via user testing feedback regarding recommendation relevance and provider clarity.